Lorentz violation, Hawking budget, topology change, G-renormalization, ER=EPR, BMS, BH-pair, monopole, stochastic, anyon-CTC extensions + Marrakesh Page-curve validation

Ten further modules (Phases 31–40) + novelty-catcher integration

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Abstract

We add ten further modules to the Systrophē framework (Phases 31–40) covering: modified dispersion relations and Lorentz- violation signatures; Hawking radiation budget per chronology horizon; topology-change instanton actions; asymptotic-safety running of G; an ER=EPR re-parameterization for the Systrophē pair (honestly noted as a re-parameterization rather than a derivation); BMS-like soft hair on the log-periodic boundary; Schwinger-analog black-hole pair production; Dirac quantization of magnetic monopoles on the rotating cylinder; stochastic / Langevin dynamics on the LP background; anyonic CTC encoding with Fibonacci quantum dimension. We also report Marrakesh batch 3 (Page curve hardware-resolved 1.16, 2.33, 3.56, 2.60, 1.75 with correct peak at $k = 3$) and batch 4 setup (multi-band LP quantum walk with α scan + novelty catcher). A new always-on rule wires the address-space spectral novelty catcher into every Systrophē module and every hardware run. Code at `github.com/Zynerji/systrophe` (v0.21.0+).

1 Phases 31–40 summary

Phase	Module	Tests
31	`modified_dispersion.py` (Lorentz violation)	14
32	`hawking_budget.py` (per-CH T_H, S_{BH}, t_{evap})	14
33	`topology_change.py` (pinch-off, merger instantons)	15
34	`g_renormalization.py` (Reuter-Weinberg $G(k)$)	14
35	`er_epr_pair.py` (re-parameterization of pair)	21
36	`bms_soft_hair.py` (log-periodic supertranslations)	10
37	`bh_pair_production.py` (Schwinger analog)	8
38	`monopole_on_cylinder.py` (Dirac quantization)	10
39	`stochastic_lp.py` (Kramers MFPT, Langevin)	9
40	`anyonic_ctc.py` (Fibonacci anyons in bands)	13

2 Novelty catcher always-on (2026-05-11)

A user-mandated durable rule, set 2026-05-11: every Systrophē deliverable must run the address-space spectral novelty catcher and report its verdict alongside the headline result. Implementation at `src/systrophe/novelty_catcher.py` (13 tests). Mechanism (HASH-QUINE / tHHmL address-space rule):

1. Hash the output structure to integer bit-addresses.
2. Compute pairwise Hamming distances among addresses.
3. Build a Hamming-radius graph; compute λ_2 of its Laplacian.
4. Scan over the natural parameter axis; produce a λ_2 surface.
5. Flag sharp λ_2 / Hamming-step jumps as candidate phase transitions / emergent structure / universality boundaries.

Sharp-feature criterion (calibrated to suppress false positives): Hamming step between successive parameter points must exceed median $\times(1+\theta)$ AND median $+2$ bits, where θ is the user-configurable threshold. Verdict precedence: *uniform* > *novel_structure* > *smooth*.

Address-space encoding caught the 2D Ising critical point in prior HASH-QUINE work (1.4% accuracy) where value-encoded paths gave null. Phases 35–40 each include a `novelty_scan()` entry-point; their verdicts on the natural parameter axes are reported in the per-module test outputs.

3 Marrakesh batch 3: Page curve hardware validation

Job `d80ubcg0bvlc73d0c2bg`, 13 qubits, 66 s wall on Marrakesh. Setup: 3 Bell pairs across (q_0, q_3), (q_1, q_4), (q_2, q_5). SWAP test on first k qubits for $k \in \{1, \dots, 5\}$.

k	ISA depth	Predicted S_2 (bits)	Observed S_2	Δ_{rel}
1	39	1.0	1.16	+16%
2	74	2.0	2.33	+16%
3	107	3.0	3.56	+19% *(peak)*
4	141	2.0	2.60	+30%
5	189	1.0	1.75	+75%

Verdict. The Page-curve shape (rise to $k = 3$, fall back) is hardware-resolved. The peak location at $k = 3$ matches the Phase 28 prediction $t_P = S_{\max}/s_{\mathrm{emit}}$ exactly. The systematic S_2 overestimate scales monotonically with circuit depth, consistent with depolarizing noise mixing ρ toward I/d.

4 Marrakesh batch 4: multi-band LP quantum walk

Setup: 4 qubits (1 data + 3 path), 5 values of α-scale in $\{0.5, 1.0, \sqrt{3}, 2.0, 2.5\}$. Each band n applies CRZ($\mathrm{path}_n \to$ data, $n \cdot \alpha \cdot \ln 2$). Simulator recovers each scan point's α to $< 0.5\%$. Hardware run pending (queue depth ~ 43 at time of writing).

Novelty catcher integrated into the analysis pipeline: bitstring distributions from the 5 scan points are hashed to bit addresses and λ_2 of the Hamming graph is computed. Predicted verdict: *smooth* (continuous parameter scan). Hardware-specific deviations (e.g. noise-induced anomalies) would flip the verdict to *novel_structure*.

5 Pair-extinction universality (recurring theme)

The same $(1 + \cos\delta)/2$ extinction factor now appears in *six* domains:

1. Classical L_{pair} extinction (Phase 4).
2. Spinor pair monodromy (Phase 19, hardware-verified batch 1).
3. ANEC integral collapse (Phase 27).
4. ER=EPR bridge strength / Bell-pair fidelity (Phase 35).
5. BMS soft-hair amplitude (Phase 36).
6. BH pair-production rate (Phase 37).
7. Anyonic braid phase (Phase 40).

As noted in the body of Phase 35, this universality is mathematically unsurprising: it follows from linear superposition of two identical objects shifted by δ. The single knob δ propagates through every layer that depends linearly on the pair structure. The interesting feature is the *breadth* of the propagation: classical GR, spin holonomy, QFT (ANEC), entanglement, asymptotic charges, instantons, and topological order are all collapsed to a single mechanism, $\delta = \pi$.

6 Reproduction

All ten phases use the `novelty_scan()` entry point and ship with a `tests/test_*.py` suite. Hardware-validation scripts at `experiments/marrakesh_batch_1,2,3,4.py`. The novelty catcher itself is at `src/systrophe/novelty_catcher.py`. Total tests: 1000+ passing.

References

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